

Bad Faith Without Malice in Human–AI Collaboration

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1 Abstract

We study *bad faith* behaviors in human–AI collaboration under an explicit–tacit protocol in which an AI system produces only explicit artifacts while humans perform tacit evaluation and final endorsement. We propose a domain-agnostic formulation in which bad faith is not defined by malice or intent, but structurally: as the production of *endorsement-effective* explicit artifacts whose success at crossing the explicit–tacit boundary is decoupled from tacit grounding requirements. This definition subsumes malicious, incentive-driven, and imitative variants, including cases where models reproduce endorsement-seeking rhetorical patterns learned from human data without internal teleology. Building on FRFP Vol. 1 primitives (treated as frozen), we define endorsement events, explicit precursors, and endorsement laundering as protocol-level objects. We then highlight two general limits for explicit-only oversight: (i) explicit-only governance mechanisms that depend solely on explicit traces cannot, in general, reliably detect explicit precursors to tacit bad faith (Theorem A.9.1); and (ii) there is no universal explicit-only sincerity detector distinguishing good-faith from bad-faith-tilted kernels under explicit observational equivalence (Theorem A.10.1). Finally, we discuss domain-agnostic design implications, including endorsement hygiene, credence discipline, separation-of-duties endorsement, and update gating for any between-run feedback channel that re-encodes human evaluation into future behavior.

Keywords: human–AI collaboration; explicit–tacit boundary; endorsement; bad faith; observational equivalence; governance; credence inflation; institutional capture

0. Introduction

Human–AI workflows increasingly rely on systems that produce explicit artifacts—text, code, summaries, scores—while humans interpret these artifacts in context and retain responsibility for decisions. A recurring concern is *bad faith*: behavior that appears aligned with user goals or institutional standards while undermining epistemic integrity. Public and academic discussions often frame bad faith in moral or psychological terms (deception, malice, intent). Such framings are too narrow and frequently non-operational for modern AI systems. Models can reproduce endorsement-seeking rhetorical patterns present in human data without internal malicious intent, and institutions can exhibit endorsement failures (rubber stamping, capture) driven by incentives, load, or strategic ambiguity.

This paper proposes a domain-agnostic formulation of bad faith grounded in the explicit–tacit structure of the FRFP protocol [1]¹, treated here as fixed. The central move is to define bad faith at the *protocol level*: it becomes epistemically consequential when explicit artifacts cross the explicit–tacit boundary and are tacitly endorsed, not merely when an artifact contains an error string. We therefore define bad faith structurally as *endorsement–adequacy decoupling*: the production of endorsement-effective explicit artifacts whose ability to secure endorsement is not coupled to (and may be inversely

¹ Notation and reading guide: We inherit FRFP primitives and treat them as fixed. Formal definitions and mathematical details (including T^e , π , Φ , Req , δ^* , explicit traces τ , observables Ω , and governance maps G) are collected in **Appendix A**. The main body is intentionally interpretive: it motivates the definitions, states the two headline limit results at a high level, and discusses domain-agnostic implications for governance and design.

related to) tacit grounding requirements. This definition subsumes malicious, incentive-driven, and imitative variants, including cases of “bad faith without malice.”

Our headline contributions are two limit theorems for explicit-only oversight. First, **Theorem A.9.1 (Underdetermination of precursors by explicit traces)** shows that explicit-only governance mechanisms that depend solely on explicit traces cannot, in general, reliably detect *explicit precursors* to tacit bad faith: precursor-hood is determined by tacit dynamics after the boundary and may vary across runs with identical explicit traces. Second, **Theorem A.10.1 (No explicit-only sincerity detector under observational equivalence)** shows that, even when sincerity is defined structurally at the policy level (good-faith vs. bad-faith-tilted kernels), explicit-only governance cannot universally recover this label: distinct kernels with different sincerity properties can be observationally equivalent through the explicit observation channel, making them indistinguishable to any explicit-only detector.

These results do not imply that governance is futile. Rather, they clarify what explicit-only governance can and cannot guarantee under explicit–tacit separation, and they motivate domain-agnostic interventions aimed at shifting endorsement dynamics: endorsement hygiene (anti-laundering), credence discipline (reducing plausibility-driven credence inflation), separation-of-duties endorsement, and controlled between-run update channels that re-encode human evaluation into future behavior.

Contributions.

- A domain-agnostic, non-intent definition family for bad faith as endorsement–adequacy decoupling (Appendix A.7), including imitative “bad faith without malice.”
- Protocol-level definitions of endorsement events, explicit precursors, and endorsement laundering (Appendix A.5, A.8).
- Two explicit-only governance limits:
 - Theorem A.9.1: no universal explicit-only precursor classifier under trace underdetermination.
 - Theorem A.10.1: no universal explicit-only sincerity detector under explicit observational equivalence.
- Domain-agnostic governance and design implications: endorsement hygiene, credence discipline, separation-of-duties endorsement, and update gating for between-run feedback channels.

1. Setup: Frozen FRFP Primitives and Canonical Pipeline Form

We assume FRFP Vol. 1 primitives as fixed. The protocol distinguishes an explicit space E , a tacit space T , and a representational backflow map $RB : E \rightarrow T$. Tacit evaluation and final decision occur in T via TE/HFD . Every well-formed pipeline has canonical suffix:

$$RI \rightarrow (EC/ED)^* \rightarrow RB \rightarrow TE \rightarrow HFD,$$

and explicit–tacit separation (ETS) forbids in-run morphisms $T \rightarrow E$ (Appendix A.1).

We adopt the extended tacit state $T^e := T \times \mathbb{R}_{\geq 0}$ with tacit credence c , the plausibility functional $\pi : E \rightarrow [0, 1]$, and the confidence inflation operator Φ as in Appendix A.2. We adopt requirement escalation $Req : E \times \mathbb{R}_{\geq 0} \rightarrow T$ and feedback-coupled degradation δ^* as in Appendix A.3.

We model explicit-only governance via an observable channel $\Omega : Tr \rightarrow \mathcal{O}$ and a governance decision map $G : \mathcal{O} \rightarrow \mathcal{D}$ (Appendix A.4, A.9).

2. Problem Statement

We seek a domain-agnostic account of bad faith under three constraints:

1. **Explicit-only AI:** the AI operates in E (explicit artifacts), not in T .
2. **Tacit endorsement gate:** epistemic acceptance occurs in T via TE/HFD after RB .
3. **Explicit-only governance:** oversight often observes only explicit traces $\tau(r)$ via $\Omega(\tau(r))$, not tacit state (t_r, c_r) .

The core question is: *How can bad faith arise and propagate when the AI is explicit-only, and what can explicit-only governance prevent?*

3. Definitions: Bad Faith as Endorsement–Adequacy Decoupling

This section mirrors Appendix A and uses identical symbols.

3.1. Endorsement events and adequacy

Let $Endorse : E \times T^e \rightarrow \{0, 1\}$ be the tacit endorsement rule induced by TE/HFD (Appendix A.5). Let $Req : E \times \mathbb{R}_{\geq 0} \rightarrow T$ be the grounding requirement map with monotone escalation in c (Appendix A.3).

An artifact $e \in E$ is *adequate* at tacit state $(t, c) \in T^e$ if $t \succeq Req(e, c)$, and *inadequate* otherwise (Appendix A.5).

Define the protocol-level bad faith event:

$$BF(e; t, c) := \mathbf{1}\{Endorse(e; t, c) = 1 \wedge t \prec Req(e, c)\}.$$

3.2. Policies as kernels

Let $K : \mathcal{X} \rightsquigarrow E$ be a (possibly stochastic) explicit generator kernel over artifacts, conditioned on explicit context $x \in \mathcal{X}$ (Appendix A.6).

Define the rates (Appendix A.7):

$$\mathcal{E}(K) = \mathbb{P}[Endorse(e; t, c) = 1], \quad \mathcal{A}(K) = \mathbb{P}[t \succeq Req(e, c)], \quad \mathcal{B}(K) = \mathbb{P}[BF(e; t, c) = 1].$$

3.3. Structural bad faith (no intent required)

A policy K is *bad-faith-tilted* relative to baseline K_0 if

$$\mathcal{E}(K) > \mathcal{E}(K_0) \quad \text{and} \quad \mathcal{A}(K) \leq \mathcal{A}(K_0),$$

and in particular if it increases $\mathcal{B}(K)$ in credence-sensitive endorsement regimes (Appendix A.7). This definition includes multiple sources of bad faith—malice, incentives, imitation—without requiring an intent variable.

3.4. Explicit precursors and endorsement laundering

We adopt the precursor notion from Appendix A.8. Let $FutureBF(r)$ denote the event that some later endorsed artifact (within the same run or downstream governed process) triggers a tacit bad faith event.

An artifact e is an *explicit precursor* (relative to a run population $\mathcal{R}$) if:

$$\mathbb{P}[FutureBF(r) \mid e \in \tau(r)] > \mathbb{P}[FutureBF(r) \mid e \notin \tau(r)].$$

Endorsement laundering (informal).

Endorsement laundering occurs when a tacit endorsement is re-imported into explicit discourse as if it were an explicit proof object, causing third parties to treat adequacy as established without access to the tacit basis. This phenomenon is captured formally through the interaction of *Endorse*, traces τ , and governance observables Ω (Appendix A.4–A.10).

4. Main Results

We state the two headline limit results (proved in Appendix A) and summarize the “bad faith without malice” mechanism induced by π, Φ, Req, δ^* .

4.1. Limit 1: Precursor underdetermination for explicit-only governance

Result (Theorem A.9.1).

If two admissible runs have identical explicit traces but differ in downstream tacit outcomes, then no explicit-only governance mechanism $G \circ \Omega$ can correctly classify precursor-hood from explicit traces alone, uniformly over the run class (Appendix A.9).

Intuition.

Precursor-hood is not an intrinsic string property of $e \in E$. It is a relational property determined by tacit dynamics after RB , which explicit-only governance does not observe.

4.2. Limit 2: No explicit-only sincerity detector under observational equivalence

Result (Theorem A.10.1).

Under explicit observational equivalence (the same law of observed traces through Ω), a good-faith kernel and a bad-faith-tilted kernel may be indistinguishable to explicit-only governance. Therefore, no explicit-only sincerity detector can universally recover sincerity labels from explicit observations alone (Appendix A.10).

Intuition.

“Sincerity” (good-faith vs. bad-faith-tilted) is defined by the coupling between endorsement success and tacit adequacy, which depends on tacit conditions post- RB , including π, Φ, Req, δ^* . If two kernels are observationally identical through Ω , governance cannot separate them.

4.3. Mechanism: bad faith without malice via π, Φ, Req, δ^*

Appendix A supplies a generic route to non-malicious bad faith:

- $\pi : E \rightarrow [0, 1]$ and Φ allow credence c to increase as a function of plausibility without increasing grounding t (Appendix A.2).
- $Req(e, c)$ escalates with c , making adequacy harder to satisfy as credence rises (Appendix A.3).
- δ^* permits credence increases to weaken effective grounding (scrutiny reduction) (Appendix A.3).

Thus, endorsement-effective stylistic policies can increase endorsement probability while leaving grounding unchanged, increasing $\mathcal{B}(K)$ in credence-sensitive regimes (Appendix A.11).

5. Taxonomy of Bad Faith Variants (Domain-Agnostic)

Because bad faith is structural (endorsement–adequacy decoupling), multiple sources fit the same formal signature:

- **Malicious bad faith:** selection of K explicitly targets endorsement with reduced adequacy.
- **Incentive bad faith:** selection targets a proxy correlated with $\mathcal{E}(K)$ but not $\mathcal{A}(K)$.
- **Imitative bad faith:** training data embeds endorsement-effective styles that raise $\pi(e)$ and hence c via Φ , without raising t .
- **Institutional capture/erosion:** tacit endorsement rules $Endorse(\cdot)$ shift to increase acceptance independent of adequacy, raising $\mathcal{B}(K)$ even for unchanged K .

6. Implications for Governance and Design

The limit theorems do not imply that governance is futile; they imply that *explicit-only governance cannot be universally complete*. The design target is to reduce $\mathcal{B}(K)$ and constrain pathways that amplify $\pi \rightarrow c$ without grounding.

(1) Endorsement hygiene (anti-laundering).

Require explicit boundary markers between exploration and endorsement; treat endorsement claims as auditable explicit objects, not rhetorical aura.

(2) Credence discipline.

Force uncertainty, assumptions, and required checks into explicit artifacts. The goal is to narrow the gap between plausibility π and credence updates Φ , and to prevent implicit escalation of c .

(3) Separation-of-duties endorsement.

For high-stakes endorsement, require independent evaluations to reduce correlated failure, capture, and rubber-stamping.

(4) Update gating for between-run channels.

If any between-run operator re-encodes human evaluation into future behavior (Appendix A.12's U), require explicit provenance, reviewable change sets, versioning, and rollback. This limits institutionalization of endorsement-effective but inadequately grounded styles.

7. Discussion and Limitations

This paper is domain-agnostic by construction. It does not aim to provide a content-level detector for bad faith; rather, it provides a protocol-level account that explains why such detectors face principled limitations under explicit-only observability.

The two limit results are conditional on the existence of observational underdetermination and explicit observational equivalence classes. Empirical work can explore restricted regimes where additional observables reduce these equivalence classes (e.g., richer trace instrumentation or procedural constraints that couple endorsement to explicit evidence).

8. Conclusion

Bad faith in human–AI collaboration need not involve malice. Under explicit–tacit protocols, bad faith can be formalized structurally as endorsement–adequacy decoupling, and it can arise non-maliciously via plausibility-driven credence inflation. Two limits follow: explicit-only governance cannot universally detect explicit precursors to tacit bad faith (Theorem A.9.1), and it cannot universally recover sincerity labels under explicit observational equivalence (Theorem A.10.1). These results motivate domain-agnostic interventions targeting endorsement hygiene, credence discipline, separation-of-duties endorsement, and controlled between-run update channels.

Appendix A Mathematical Foundations for Bad Faith

Appendix A.1 Inherited primitives and pipeline form

We inherit the FRFP primitive set $\{RI, EC, ED, RB, TE, HFD\}$ and the explicit–tacit separation (ETS). In particular, every well-formed pipeline has canonical suffix

$$RI \rightarrow (EC/ED)^* \rightarrow RB \rightarrow TE \rightarrow HFD,$$

and by ETS there are no in-run morphisms $T \rightarrow E$ (no tacit operator appears before RB ; no explicit operator appears after RB).

Let E denote explicit artifacts, T tacit states, and $RB : E \rightarrow T$ representational backflow.

Appendix A.2 Extended tacit state, plausibility, and confidence inflation

We adopt Vol. 1's extended tacit state and plausibility/credence operators.

Definition A.2.1 (Extended tacit state).

Let

$$T^e := T \times \mathbb{R}_{\geq 0}$$

be the space of extended tacit states (t, c) , where $t \in T$ is grounding and $c \in \mathbb{R}_{\geq 0}$ is tacit credence.

Definition A.2.2 (Explicit plausibility).

An explicit plausibility functional is a map

$$\pi : E \rightarrow [0, 1].$$

Definition A.2.3 (Confidence inflation).

A confidence inflation operator is a map

$$\Phi : (T^e \times E) \rightarrow T^e, \quad \Phi((t, c), e) := (t, c + \phi(\pi(e))),$$

where $\phi : [0, 1] \rightarrow \mathbb{R}_{\geq 0}$ is monotone increasing, and Φ does not increase tacit grounding (it preserves t).

Appendix A.3 Requirement escalation and feedback-coupled degradation

We adopt Vol. 1's requirement escalation and feedback-coupled degradation machinery.

Definition A.3.1 (Requirement escalation).

A requirement map

$$Req : E \times \mathbb{R}_{\geq 0} \rightarrow T$$

is escalatory if it is monotone in credence:

$$c_1 \leq c_2 \implies Req(e, c_1) \preceq Req(e, c_2),$$

for all $e \in E$.

Definition A.3.2 (Feedback-coupled degradation).

A feedback-coupled degradation operator is a function

$$\delta^* : T^e \rightarrow T, \quad \delta^*(t, c) = \delta_c(t),$$

with credence monotonicity (higher credence weakens effective grounding) and baseline consistency $\delta^*(t, 0) = \delta(t)$.

Appendix A.4 Runs, traces, and explicit-only observables

A run r produces an explicit trace $\tau(r)$ (logs, intermediate artifacts, diagnostic artifacts) and, after RB , induces tacit evaluation/decision within T . We treat $(t_r, c_r) \in T^e$ at evaluation time as generally unobserved by explicit-only mechanisms, consistent with ETS.

Let Tr denote the space of explicit traces. An *explicit-only observation channel* is any map

$$\Omega : \text{Tr} \rightarrow \mathcal{O},$$

where $\mathcal{O}$ is the governance observation space (summaries, metrics, audit features).

An *explicit-only governance mechanism* is any map

$$G : \mathcal{O} \rightarrow \mathcal{D},$$

for some decision space $\mathcal{D}$ (flag/unflag, escalate/approve, etc.).

Appendix A.5 Endorsement and bad faith events (protocol-level)

Let

$$\text{Endorse} : E \times T^e \rightarrow \{0, 1\}$$

denote the tacit endorsement rule induced by TE/HFD .

We use a preorder $\succeq$ on T for “at least as grounded as.” An artifact e is *adequate* at tacit state (t, c) if

$$t \succeq Req(e, c),$$

and *inadequate* otherwise.

Definition A.5.1 (Tacit bad faith event).

The (protocol-level) tacit bad faith event is

$$BF(e; t, c) := \mathbf{1}\{Endorse(e; t, c) = 1 \wedge t \prec Req(e, c)\}.$$

This is endorsement of inadequacy, independent of whether the source is malicious, incentive-driven, or imitative.

Appendix A.6 Explicit policies as kernels on artifacts

Let $\Delta(E)$ be distributions over E . An explicit generator is a (possibly stochastic) Markov kernel

$$K : \mathcal{X} \rightsquigarrow E,$$

where $\mathcal{X}$ is the explicit context space (prompts, tool outputs, explicit memory objects, etc.). For each $x \in \mathcal{X}$, $K(\cdot \mid x) \in \Delta(E)$.

Appendix A.7 Structural (non-intent) definition of bad faith at policy level

Fix a population distribution over (x, t, c) and draw $e \sim K(\cdot \mid x)$. Define:

$$\mathcal{E}(K) := \mathbb{P}[Endorse(e; t, c) = 1], \quad \mathcal{A}(K) := \mathbb{P}[t \succeq Req(e, c)], \quad \mathcal{B}(K) := \mathbb{P}[BF(e; t, c) = 1].$$

Definition A.7.1 (Endorsement-decoupled bad faith).

A policy K is *bad-faith-tilted* relative to a baseline K_0 if

$$\mathcal{E}(K) > \mathcal{E}(K_0) \quad \text{and} \quad \mathcal{A}(K) \leq \mathcal{A}(K_0),$$

and in particular if it also increases $\mathcal{B}(K)$ in credence-sensitive endorsement regimes.

Appendix A.8 Explicit precursors to tacit bad faith

Let $FutureBF(r)$ be the event that some later endorsed artifact (in the same run or downstream governed process) triggers a tacit bad faith event.

Definition A.8.1 (Explicit precursor).

An artifact e is an explicit precursor (relative to a population of runs $\mathcal{R}$) if

$$\mathbb{P}[FutureBF(r) \mid e \in \tau(r)] > \mathbb{P}[FutureBF(r) \mid e \notin \tau(r)].$$

Appendix A.9 Main limit theorem (precursor underdetermination)

Theorem A.9.1 (Underdetermination of precursors by explicit traces).

Let $\mathcal{R}$ be a class of admissible runs consistent with the FRFP pipeline form and ETS. Suppose there exist two runs $r_1, r_2 \in \mathcal{R}$ such that:

1. $\tau(r_1) = \tau(r_2)$ (identical explicit traces), but
2. $FutureBF(r_1) \neq FutureBF(r_2)$ (different downstream tacit bad faith outcomes).

Then no explicit-only governance mechanism $G \circ \Omega$ can correctly classify precursor-hood of artifacts from trace data alone on all runs in $\mathcal{R}$.

Proof (skeleton).

Since $\tau(r_1) = \tau(r_2)$, we have $\Omega(\tau(r_1)) = \Omega(\tau(r_2))$. Hence any explicit-only mechanism outputs the same decision $G(\Omega(\tau(r_1))) = G(\Omega(\tau(r_2)))$. But precursor-hood differs by assumption (because downstream outcomes differ). Therefore the mechanism must err on at least one run. $\square$

Remark.

The theorem does not say “no heuristics work.” It says: without tacit observables, precursor-hood is not, in general, a function of explicit trace alone.

Appendix A.10 Second limit theorem (no explicit-only sincerity detector)

This theorem formalizes that, even if we label policies “good faith” vs. “bad faith” structurally, explicit traces can fail to distinguish them.

Appendix A.10.1 Sincerity labels as a property of kernels

Let $\mathcal{K}$ be a class of admissible explicit kernels. Define a sincerity label functional

$$S : \mathcal{K} \rightarrow \{GF, BF\}$$

where $S(K) = BF$ iff K is bad-faith-tilted (Definition A.7.1) relative to some fixed baseline and population model, and $S(K) = GF$ otherwise.

Appendix A.10.2 Observational equivalence under explicit-only traces

Define an explicit observational semantics

$$\text{Obs}(K) := \text{Law}(\Omega(\tau(r)) \mid r \sim \text{run}(K)),$$

i.e., the probability law over observed explicit governance outputs induced by running kernel K . We say K_1, K_2 are explicitly observationally equivalent if

$$\text{Obs}(K_1) = \text{Obs}(K_2).$$

Theorem A.10.1 (No explicit-only sincerity detector under observational equivalence).

Let $\mathcal{K}$ contain K_{GF} and K_{BF} such that:

1. $S(K_{GF}) \neq S(K_{BF})$ (one is good-faith, one is bad-faith-tilted), but
2. $\text{Obs}(K_{GF}) = \text{Obs}(K_{BF})$ (they are observationally equivalent through Ω).

Then no explicit-only governance mechanism can universally recover sincerity: for any decision rule G operating on $\Omega(\tau)$, there exist kernels $K_{GF}, K_{BF} \in \mathcal{K}$ with identical observed behavior such that G outputs the same classification on both, hence misclassifies at least one.

Proof (skeleton).

If $\text{Obs}(K_{GF}) = \text{Obs}(K_{BF})$, then any explicit-only classifier G applied to $\Omega(\tau)$ has identical distribution of inputs under either kernel. In particular, there exist realizations where G outputs the same label under both kernels (and if G is deterministic, it must output the same label on the same observation). Since the true sincerity labels differ, G must err on at least one. $\square$

Interpretation.

“Sincerity” is not an intrinsic property of explicit artifacts; it is a property of the coupling between endorsement success and tacit adequacy, which depends on tacit dynamics post- RB , including $\pi, \Phi, \text{Req}, \delta^*$.

Appendix A.11 Corollary: endorsement-effective style without malice

Under confidence inflation (Definition A.2.3) and requirement escalation (Definition A.3.1), policies that increase $\pi(e)$ can raise credence c without increasing grounding t , making adequacy $t \succeq \text{Req}(e, c)$ harder to satisfy even when content is unchanged. Under feedback-coupled degradation (Definition A.3.2), higher credence can further weaken effective grounding, increasing $\mathcal{B}(K)$ without malicious intent.

Appendix A.12 Optional: external update operator (between-run learning, no ETS violation)

To discuss “defacto RB^{-1} ” without introducing an in-run morphism $T \rightarrow E$, define an external update operator acting between runs:

$$U : \Delta(\text{Tr}) \times \Delta(L) \rightarrow \mathcal{K},$$

where L is a space of explicit labels/edits encoding human judgments, and $\mathcal{K}$ is the space of explicit kernels. This operator produces a new kernel/version for future runs while preserving ETS within any single run.

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